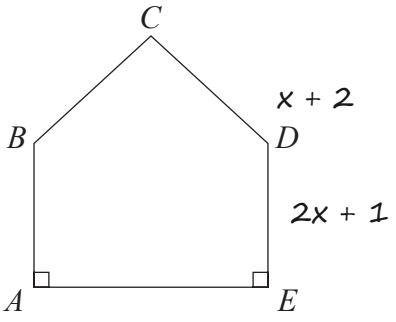
17 The diagram shows a pentagon.
The pentagon has one line of symmetry.



$AE = 4x$
$AB = 2x + 1$
$BC = x + 2$

All these measurements are given in centimetres.

The perimeter of the pentagon is 18 cm.

(a) Show that $10x + 6 = 18$

By symmetry, DE = AB and CD = BC.

1 mark
Identify the two matching sides

Perimeter = 4x + (2x+1) + (x+2) + (x+2) + (2x+1)

1 mark
Include all five sides

= (4+2+1+1+2)x + (1+2+2+1) = 10x + 6

The perimeter is 18, so 10x + 6 = 18.

1 mark
Simplify fully and equate to 18 **(3)**

(b) Find the value of x.

10x + 6 = 18
(-6) (-6)
10x = 12
(÷10) (÷10)
x = 1.2

1 mark
Subtract 6 from both sides

1 mark
Correct value of x

$x =$ 1.2

(2)

(Total for Question 17 is 5 marks)